BRD — Offline Bangla Dubbing Video Player

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A cross-platform desktop video player that dubs any spoken language into Bangladeshi Bangla voice, offline, with a distinct voice per speaker.

Business problem

Bangla speakers who do not understand the original language cannot follow most foreign films, lectures and shows. Subtitles demand constant reading, which excludes children, elderly viewers, people with low literacy or poor eyesight, and anyone multitasking.

Professional Bangla dubbing is expensive, slow, and only exists for a small share of content. Existing AI dubbing services are online-only, paid per minute, and send private videos to third-party servers. Few of them offer Bangladeshi Bangla at all.

The gap: no offline player turns any video into spoken Bangladeshi Bangla on the user's own computer.

Objectives and success metrics

The proposed targets below are starting values to confirm during gap analysis.

Objective	Success metric (proposed target)
Understandable Bangla dubbing	80%+ of test viewers follow the story without subtitles
Correct speaker voices	Speaker labels correct in 90%+ of lines with 2–4 speakers
Fully offline	Zero network calls after first-run model download
Fast enough to use	Pre-process: under 15 min per 1 hour of video on an 8 GB NVIDIA GPU; live mode starts within 60 s
Stays in sync	Dubbed line starts within 0.5 s of the original line
Cross-platform	Installers for Windows, macOS and Linux pass the same test set

Scope

In scope (version 1)

- Desktop player built with PyQt for Windows, macOS and Linux
- Common video formats (MP4, MKV, AVI, MOV, WebM) through ffmpeg
- Automatic source-language detection and speech-to-text
- Translation into Bangla
- Bangla text-to-speech with a distinct voice per detected speaker
- Original music and sound effects kept under the dubbed voices
- Pre-process mode and buffered live mode
- Export of the dubbed audio track or a dubbed video file
- Optional Bangla subtitle display from the same translation

Out of scope (version 1)

- Lip-sync or any change to the video picture
- Cloning each original actor's voice into Bangla
- Streaming sites, YouTube URLs or live TV input
- Translation into languages other than Bangla
- Mobile apps and cloud processing

Stakeholders and target users

Stakeholder	Role / interest
Product owner	Owns the requirements, approves UAT and release
Development team	Builds the PyQt app and the AI pipeline
QA team	Tests dubbing quality, sync and each platform
UAT testers	Native Bangladeshi Bangla speakers who judge naturalness
End users: home viewers	Watch foreign films and shows in Bangla, including children and elderly family members
End users: students	Follow foreign-language lectures and tutorials

Stakeholder	Role / interest
End users: low-connectivity areas	Need everything to work without internet

Names for each internal role are still to be assigned.

Business requirements

Priority uses MoSCoW: Must, Should, Could.

ID	Requirement	Priority
BR-01	The user opens a local video file and plays it with standard controls (play, pause, seek, volume, fullscreen).	Must
BR-02	The system detects the spoken language automatically; the user can override it.	Must
BR-03	The system replaces the original speech with spoken Bangla, not subtitles.	Must
BR-04	The Bangla speech uses Bangladeshi vocabulary and pronunciation where models allow.	Must
BR-05	Each detected speaker gets a different, consistent Bangla voice for the whole video.	Must
BR-06	The system keeps original music and sound effects under the dubbed voices.	Must
BR-07	All processing runs on the user's computer with no internet after setup.	Must
BR-08	Pre-process mode dubs the whole video before playback, perfectly synced.	Must
BR-09	Buffered live mode starts playback within 60 s and keeps ahead of playback.	Should
BR-10	The user can choose or change the voice assigned to each speaker.	Should

ID	Requirement	Priority
BR-11	The user can switch between original and Bangla audio during playback.	Should
BR-12	The user can export the dubbed audio or a dubbed video file.	Should
BR-13	The user can show Bangla subtitles from the same translation.	Could
BR-14	Processed videos are cached so replaying needs no reprocessing.	Could
BR-15	The user can review and edit the Bangla text of a line, then regenerate its voice.	Could
BR-16	The app interface is available in Bangla and English.	Could

Non-functional requirements

ID	Area	Requirement
NFR-01	Platforms	Windows 10/11, macOS 12+ (Apple Silicon first), Ubuntu 22.04+
NFR-02	Hardware	Recommended: NVIDIA GPU with 8 GB+ VRAM or Apple Silicon with 16 GB RAM. Minimum: CPU only, pre-process mode only
NFR-03	Privacy	No video, audio or text leaves the device
NFR-04	Storage	Models take about 5–10 GB, downloaded on first launch with a progress bar and resume
NFR-05	Sync	Dubbed lines start within 0.5 s of the original
NFR-06	Reliability	A failed line falls back to the original audio instead of stopping playback
NFR-07	Usability	A first-time user dubs a video in 3 clicks or fewer
NFR-08	Licensing	Every model and library is licensed for the intended use (commercial or non-commercial, to confirm)

ID	Area	Requirement
NFR-09	Packaging	One installer per platform (PyInstaller or Nuitka)

Assumptions, constraints and risks

Assumptions

- Users accept a one-time model download before going fully offline.
- "Instant" means a short buffer (30–60 s) or a pre-process step, not zero delay.
- Distinct preset Bangla voices per speaker are acceptable in place of cloned actor voices.

Constraints

- Built in Python with PyQt.
- Uses open-source models only, to stay offline and free of per-minute fees.

Risks

Risk	Impact	Mitigation
Open Bengali voices sound Kolkata, not Bangladeshi	High	Fine-tune a TTS model on Bangladeshi recordings; evaluate with Bangladeshi UAT testers
Few good Bangla voices limit multi-speaker variety	Medium	Combine voice presets with pitch and speed variation
Translation sounds literal or unnatural	Medium	Compare NLLB, IndicTrans2 and a local LLM in the spike
Live mode too slow on mid-range PCs	Medium	Default to pre-process mode on weak hardware
Model licenses block commercial release	High	License review before development starts
Large app and model size	Low	Download models on first run; offer smaller models

Open questions for gap analysis

- [ ] Is the product commercial, free, or internal? This decides which model licenses are allowed.
- [ ] Which source languages matter most first (English, Hindi, Korean, Arabic, others)?
- [ ] What is the minimum hardware the target users actually own?
- [ ] Is a Bangladeshi voice dataset available, or does one need to be recorded?
- [ ] Are a 30–60 s live buffer and preset voices acceptable to the product owner?
- [ ] What are the budget and target release date?
- [ ] Who are the product owner, developers, QA and UAT testers?